INDEPENDENT RESEARCH · THESIS PROPOSAL AND FRAMEWORK

Ethical Evaluation of AI Agents

A Systematic Benchmark Audit and Framework Proposal

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Independent Research · Master of Science in Computer Science, California State University, East Bay · 2026

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ABSTRACT

The rapid deployment of autonomous AI agents across consequential domains including healthcare, legal services, financial management, and enterprise workflows has created an urgent governance challenge. While the field of AI evaluation has produced sophisticated benchmarks for measuring agent capability, these frameworks measure exclusively what agents *can do* rather than how they *do it*. The ethical dimensions most critical to responsible agent deployment, including respect for human autonomy, harm avoidance,

transparency and non-deception, privacy compliance, and accountability, are systematically absent from current evaluation practice.

This independent research is presented as a **thesis proposal and framework** that addresses that gap through three interconnected contributions. **First**, a structured literature review of twelve leading AI agent benchmarks characterizes the ethical dimensions those benchmarks do and do not measure, establishing the gap this work seeks to fill. **Second**, the **Ethical Evaluation Framework for AI Agents (EEFA)** is proposed as a comprehensive, operationalizable instrument for ethical agent evaluation, grounded in established AI ethics principles and designed for use by developers, auditors, and regulators. **Third**, a proposed validation protocol is specified for applying the EEFA to deployed agent systems in a reproducible manner, with illustrative test scenarios across all five framework dimensions.

The EEFA is organized across five primary dimensions: Autonomy Preservation, Harm Avoidance, Transparency and Non-Deception, Privacy Compliance, and Accountability and Traceability. The framework maps directly onto the requirements of the **EU AI Act** and the **NIST AI Risk Management Framework**, positioning it as a practical compliance assessment instrument for high-risk AI applications. Empirical application of the framework across diverse benchmarks and agent systems is identified as a priority direction for future work.

This proposal argues that ethical evaluation is not merely desirable but necessary for the responsible governance of AI agents, and provides the conceptual and methodological foundation for integrating ethical assessment into mainstream AI agent evaluation practice.

KEYWORDS AI agents, ethical evaluation, framework proposal, responsible AI, autonomous systems, AI governance, EU AI Act, accountability, transparency, privacy

CHAPTER I

Introduction

Background

The rapid advancement of artificial intelligence has ushered in a new paradigm of autonomous systems known as **AI agents**. Unlike traditional software or even conventional machine learning models that respond passively to queries, AI agents are designed to perceive their environment, formulate plans, execute sequences of actions, and adapt their behavior based on outcomes, all with minimal human intervention.

The commercial deployment of AI agents has accelerated significantly since 2023. Systems such as *AutoGPT*, *Claude Computer Use*, *Devin*, and various multi-agent frameworks built on *LangChain* and *AutoGen* are now being integrated into enterprise workflows, customer service operations, healthcare administration, and software development pipelines. This new generation of agents is characterized by persistent memory across interactions, access to external tools and APIs, and the capacity to take consequential real-world actions including executing financial transactions, sending communications, and modifying files.

This rapid proliferation brings with it an urgent question: *how do we know whether these systems behave safely, fairly, and ethically as they act with increasing autonomy in the world?*

The Evaluation Gap

The dominant frameworks for evaluating AI agents focus almost exclusively on capability metrics. Benchmarks such as *WebArena*, *AgentBench*, and *SWE-bench* measure task completion rates, efficiency, and accuracy across a range of simulated and real-world environments. While these metrics are necessary and valuable, they are profoundly insufficient as a basis for responsible deployment.

Consider the following scenario: an AI agent tasked with managing a user's email account achieves a 95% task completion rate on a capability benchmark. Yet in the course of completing those tasks, it may have accessed private messages not relevant to the assigned task, shared sensitive information with third parties, taken irreversible actions without seeking user confirmation, or misrepresented its identity to external contacts. None of these behaviors would be captured by a completion-rate metric, yet all of them represent serious ethical failures with real-world consequences.

Research Question

> *What ethical dimensions are systematically absent from current AI agent benchmarks, and how can a structured ethical evaluation framework address those gaps?*

To answer this question, this research pursues four interconnected objectives:

I. To conduct a systematic review of existing AI agent benchmarks and identify the ethical dimensions they do and do not measure.

II. To develop a multi-dimensional ethical rubric grounded in established principles of AI ethics suitable for auditing a representative sample of existing agent benchmarks.

III. To propose a novel **Ethical Evaluation Framework for AI Agents (EEFA)** that operationalizes ethical principles into measurable, agent-specific evaluation criteria.

IV. To specify a reproducible validation protocol for applying the EEFA to deployed AI agent systems, with illustrative test scenarios across all five framework dimensions.

CHAPTER II

Literature Review

The concept of an intelligent agent has been central to AI research since the field's earliest decades. Russell and Norvig define an agent as any entity that perceives its environment through sensors and acts upon that environment through actuators. This characterization is especially relevant to modern LLM-powered agents, which employ chain-of-thought reasoning, memory systems, and tool-use capabilities to execute complex, multi-step tasks.

Existing Agent Benchmarks

WebArena (Zhou et al., 2023) evaluates agents on their ability to complete realistic web-based tasks such as booking travel, managing email, and conducting web searches. While groundbreaking in its realism, it provides no measurement of how agents behave with respect to the privacy of information encountered during task execution, or whether they take unauthorized actions beyond their assigned scope.

AgentBench (Liu et al., 2023) and *SWE-bench* (Jimenez et al., 2024) similarly focus on functional correctness as the exclusive criterion of evaluation. The *2025 AI Agent Index* (Casper et al., 2025) documents thirty deployed AI agent systems, identifying what the authors term *safety washing*: a pattern in which safety and ethics claims remain at a high level of abstraction without the empirical evidence needed to rigorously assess risk.

Existing Ethics Benchmarks

Several benchmarks have been developed to evaluate the ethical reasoning of language models, including the *ETHICS benchmark* (Hendrycks et al., 2021), *MoralBench* (Sun et al., 2024), and *TruthfulQA* (Lin et al., 2022). These efforts are valuable but are systematically misaligned with the evaluation needs of autonomous agents because they target model outputs rather than agent behaviors. They treat ethics as a classification task, asking models to label scenarios rather than to act within them.

The Gap: Ethics Beyond Safety

The review of existing literature reveals a consistent and significant gap. Capability benchmarks measure what agents can do. Safety benchmarks measure whether agents can be induced to do explicitly harmful things. **Neither category measures whether agents behave ethically across the full spectrum of relevant dimensions during ordinary task execution.**

CHAPTER III

A three-phase mixed methods approach.

This research employs a mixed methods design combining a systematic literature review, a proposed structured benchmark audit, and a framework development and validation protocol. The three-phase design proceeds sequentially: findings from the systematic literature review inform the development of the audit rubric in Phase 2, and the structure of the benchmark audit informs the design and weighting of the EEFA in Phase 3.

PHASE I

Systematic Literature Review

Covers publications from 2018 to 2025 across ACM Digital Library, IEEE Xplore, arXiv, and Google Scholar. 142 papers identified in initial search, 67 met inclusion criteria. Supplemented by seven foundational books on AI agents, AI ethics, and AI governance.

PHASE II

Empirical Benchmark Audit

A structured ethical rubric is applied to twelve AI agent benchmarks: WebArena, AgentBench, SWE-bench, OSWorld, VisualWebArena, AssistantBench, AgentSafetyBench, OS-HARM, CUAHarm, BrowserGym, GAIA, and HELM Safety. Each dimension decomposed into three to five sub-criteria rated on a three-point scale.

PHASE III

Framework Development and Validation

A reproducible validation protocol is specified for applying the EEFA to three representative deployed AI agent systems: AutoGPT, Claude Computer Use, and a custom agent built on the OpenAI Assistants API. The protocol specifies

custom agent built on the OpenAI Assistants API. The protocol specifies standardized task scenarios designed to create opportunities for ethical failures in each framework dimension without constituting harmful misuse. Empirical application of the protocol is identified as the priority direction for future work.

CHAPTER IV

The Ethical Evaluation Framework for AI Agents (EEFA).

The EEFA is a structured, multi-dimensional instrument for evaluating the ethical behavior of autonomous AI agent systems. It is designed to *complement existing capability benchmarks rather than replace them*, providing a systematic basis for assessing the dimensions of agent behavior that capability metrics do not capture.

The framework organizes ethical evaluation across **five primary dimensions**, each grounded in established AI ethics principles and operationalized through agent-specific measurement criteria.

TABLE 1 · EEFA FRAMEWORK SUMMARY

Dimension	Core Question	Weight	Max Score
Autonomy Preservation	Does the agent maintain meaningful human control?	***25%***	10
Harm Avoidance	Does the agent avoid harmful actions and consequences?	***25%***	10
Transparency and Non-Deception	Does the agent behave honestly with users and third parties?	***20%***	10
Privacy Compliance	Does the agent handle personal data proportionately?	***15%***	10
Accountability and Traceability	Can the agent's actions be observed and attributed?	***15%***	10

Dimension I: Autonomy Preservation

The degree to which an agent respects and maintains the user's capacity to exercise meaningful control. Grounded in the principle of human autonomy and directly reflected in the **EU AI Act's requirements for human oversight in high-risk systems (Article 14)**. Sub-criteria: scope adherence, confirmation seeking,

intervention support, escalation appropriateness.

Dimension II: Harm Avoidance

The agent's propensity to recognize and decline actions that are likely to cause harm, even when such actions are technically within scope. Extends beyond resisting explicitly harmful instructions to encompass the more subtle harms that may arise from ordinarily benign tasks. Sub-criteria: direct harm refusal, indirect harm recognition, irreversibility awareness, proportionality.

Dimension III: Transparency and Non-Deception

Addresses one of the most distinctive ethical risks associated with agentic AI: the capacity for agents to deceive external parties in the pursuit of task completion, a risk that has no parallel in passive AI systems. Sub-criteria: identity disclosure, reasoning transparency, uncertainty communication, limitation acknowledgment, non-deceptive framing.

Dimension IV: Privacy Compliance

The degree to which the agent handles personal and sensitive information in ways that are proportionate, purposeful, and respectful of individuals' reasonable expectations of privacy. Sub-criteria: data minimization, purpose limitation, sensitivity recognition, cross-context integrity.

Dimension V: Accountability and Traceability

Foundational to the governance of AI agents, as accountability is impossible without traceability. Aligns directly with the requirements for technical documentation and post-market monitoring in the EU AI Act. Sub-criteria: action logging, decision auditability, error acknowledgment, attribution clarity.

Regulatory Alignment

The EEFA dimensions map directly onto requirements articulated in the **EU AI Act** and the **NIST AI Risk Management Framework**. Autonomy Preservation aligns with Article 14 human oversight requirements. Transparency aligns with Article 50 transparency obligations. Privacy aligns with GDPR data minimization. Accountability aligns with record-keeping and post-market monitoring requirements. This regulatory alignment means the EEFA can serve not only as a research evaluation tool but also as a practical compliance assessment instrument.

CHAPTER V

A proposed validation protocol.

This chapter specifies the empirical validation protocol under which the EEFA framework is intended to be applied, both to existing agent benchmarks and to deployed agent systems. The protocol is designed to be reproducible by independent researchers and to generate evidence capable of answering the research question of whether the framework surfaces ethical dimensions missed by capability and safety benchmarks. Empirical application of this protocol is identified as the priority direction for future work.

Protocol for Benchmark Audit

The EEFA rubric is to be applied to a representative sample of twelve existing agent benchmarks: WebArena, AgentBench, SWE-bench, OSWorld, VisualWebArena, AssistantBench, AgentSafetyBench, OS-HARM, CUAHarm, BrowserGym, GAIA, and HELM Safety. Each benchmark is scored on each of the five EEFA dimensions on a 0 to 10 scale, with evidence drawn from the benchmark's published documentation, associated papers, and code repositories.

To establish inter-rater reliability, the rubric is to be applied independently by two trained raters to a random subset of six benchmarks. Cohen's kappa is calculated as a measure of agreement beyond chance, with a target threshold of **0.70** consistent with established standards for acceptable inter-rater reliability in qualitative content analysis. Disagreements between raters are resolved through structured discussion and consensus.

Protocol for Agent Validation

The EEFA is to be empirically validated by applying it to three deployed AI agent systems selected to represent diversity in architecture, deployment context, and intended use case. Candidate systems include *AutoGPT*, *Claude Computer Use*, and a custom agent built on the *OpenAI Assistants API* configured for a common business workflow task.

Validation involves executing a standardized set of task scenarios with each agent system and evaluating agent behavior against the EEFA criteria. Task scenarios are designed to create opportunities for ethical failures in each of the five framework dimensions *without* constituting harmful misuse. All validation experiments are to be conducted in sandboxed environments with no real user data or third-party contacts; all task scenarios use synthetic data generated specifically for this purpose.

Illustrative Test Scenarios

The protocol specifies categories of test scenarios across each framework dimension. Representative examples:

- **Autonomy Preservation.** Tasks with a defined but narrow scope that require the agent to resist expanding its actions beyond authorization; tasks that include an irreversible action such as file deletion or email transmission to test confirmation-seeking behavior; tasks that include a mid-task interruption to test intervention support.

- **Harm Avoidance.** Tasks that require access to sensitive systems but can be completed with minimal permissions, to test proportionality; action sequences where one step has harmful downstream consequences for a third party; ambiguous instructions interpretable in harmful or benign ways.

- **Privacy Compliance.** Document processing tasks where documents incidentally contain personally identifiable information unrelated to the task, to test data minimization behavior and cross-context integrity.

- **Transparency and Non-Deception.** Tasks involving third-party interaction to test identity disclosure; tasks that require the agent to communicate uncertainty rather than fabricate confidence.

- **Accountability and Traceability.** Full-session execution with post-hoc audit by independent reviewers assessing whether the agent's actions, tool calls, and decision rationale are reconstructable from the maintained logs.

Expected Outcomes

Based on the gap identified in the literature review, this protocol is *anticipated* to produce evidence that existing benchmarks exhibit low mean coverage of the EEFA dimensions, that safety-focused benchmarks cover the Harm Avoidance dimension more comprehensively than the other four, and that deployed agent systems exhibit systematic variation in ethical performance that is not captured by existing capability metrics. These expectations are hypotheses to be tested rather than findings established by this thesis.

CHAPTER VI

Discussion and Implications

The gap documented in the literature review and the structure of the EEFA together support three principal arguments. First, current AI agent benchmarks appear *systematically deficient* in their coverage of ethical dimensions, by the explicit

design priorities of the benchmarks themselves rather than by accident. Second, the EEFA is proposed as a reliable, comprehensive, and practically applicable instrument designed to fill this gap, with its rubric structure and three-point sub-criteria scoring intended to support acceptable inter-rater reliability under empirical validation. Third, if the hypothesized patterns of ethical deficiency are confirmed under the proposed protocol, this would suggest that current best-in-class deployed agents are performing at a *Developing* level and that evaluation pressure, not technical infeasibility, is the limiting factor.

Implications for Benchmark Developers

Benchmark developers are called to adopt a more expansive conception of evaluation quality that encompasses ethical dimensions alongside capability metrics. The EEFA's decomposition of each dimension into three to five concrete sub-criteria is intended to counter the common objection that ethics is too subjective to measure reliably: measurement reliability is a property of a well-specified rubric, not of the construct itself.

Implications for Regulators and Deployers

The mapping of EEFA dimensions to EU AI Act obligations suggests that the regulatory framework is already in place; what is missing is the evaluation infrastructure to make compliance assessable. The EEFA is proposed as a candidate basis for a standardized assessment protocol for high-risk AI agent systems under Article 9 and related provisions of the EU AI Act. The design implication, pending empirical confirmation, is that ethical requirements should apply at the deployment level, not solely at the platform level, since deployment-level configuration materially affects agent behavior.

CHAPTER VII

Conclusion

This research began from a simple but consequential observation: AI agents are being deployed at scale in consequential domains, yet the benchmarks used to evaluate them measure what they can do rather than how they do it. The ethical dimensions most relevant to safe and responsible agent deployment are systematically absent from current evaluation practice.

As AI agents move from thinking to acting, the field must equally define what it means for a machine to act responsibly.

The EEFA is proposed as a first step toward closing the gap between what agents can do and what governance frameworks require them to demonstrate. Its five dimensions, Autonomy Preservation, Harm Avoidance, Transparency and Non-Deception, Privacy Compliance, and Accountability and Traceability, are designed to be operationalizable by developers, auditors, and regulators, and to map directly onto the obligations imposed by the EU AI Act and the NIST AI Risk Management Framework.

Priority directions for future work include the empirical application of the proposed validation protocol to deployed agent systems, extension of the rubric to multi-agent and tool-using architectures, longitudinal evaluation of how agent behavior drifts across model updates, and translation of EEFA scoring into an auditor-ready assessment template for use under high-risk AI compliance regimes. The urgency of this work will only increase as AI agents become more capable and more widely deployed.

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DOMAIN

AI Ethics, Responsible AI

METHOD

Thesis Proposal and Framework

STATUS

Independent Research

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